In case (i), integrating ω (equivalently T) produces a functional delta $\delta[\mathcal{T}]$; in case (ii), the constraint is exact already at the classical level. In both realizations, keeping f_ϵ^2 finite leaves a standard $(\partial\epsilon)^2$ spectator; *the spurion posture* used in the main text corresponds to taking $f_\epsilon^2 \to 0^+$ (or treating ϵ as non-dynamical) *within the two-derivative truncation*, so that observables depend only on $\partial\epsilon$ through the invariant $\mathcal{T}_\mu$ (and hence through Σ_ϵ once C1 is enforced).

Low-energy effective action (power counting).

At energies $E \ll m_T$, one obtains, after eliminating ω,

$$S_{\rm eff}^{(\rm LO)} = \int d^4x\sqrt{-g}\Big[\tfrac{M_P^2}{2}R(\mathring{\omega}) + \alpha_1\, I_T + \alpha_2\,\Sigma_\epsilon\Big] \;+\; \mathcal{O}\!\left(\frac{\partial^4}{m_T^2}\right), \qquad I_T \equiv -\tfrac{1}{4}\,\Pi_{\rm PT}[T^A{}_{BC}T_A{}^{BC}], \tag{A70}$$

and *after* enforcing C1 (Section 4), $I_T = -6\eta^2\,\Sigma_\epsilon$, so that the LO bulk reduces to a single invariant line in the $\{I_T, \Sigma_\epsilon\}$ plane (Section 3.3).

Appendix G.2. Complementary EFT Pictures (Same Gradient-Only Physics)

Two complementary constructions emphasize that *only* $\partial\epsilon$ can enter observables at this order.

(i) Torsional–axion picture.

Augment (A66) by a topological coupling

$$S_{\text{t-ax}} = \int d^4x\sqrt{-g}\,\frac{\alpha}{4}\,\epsilon\,\mathrm{NY}, \qquad \mathrm{NY} \equiv d\Big(e^A \wedge T_A\Big), \tag{A71}$$

which is exact. Under A5, NY is a *boundary convention*; integrating by parts and using (A68) gives bulk terms proportional to $\partial_\mu\epsilon$ contracted with improvement currents and, hence, no new bulk Euler–Lagrange content in the PT-even sector. The low-energy reduction again depends only on Σ_ϵ (plus boundary assignments absorbed into A5).

(ii) Axion electrodynamics-like geometry.

The situation mirrors $\theta F\tilde{F}$ in gauge theory: the integrated density is topological, and only $\partial_\mu\theta$ is physically measurable (through Chern–Simons currents) at two derivatives. Replacing $(\theta, F\tilde{F})$ by (ϵ, NY) and using the scalar PT projector (A2) reproduces the same *gradient-only* observability for ϵ.

Appendix G.3. UV Hints and Emergence

While our analysis is agnostic about ultraviolet completions, two standard paths naturally generate a field with the required properties:

1. **Torsionful connections in string-inspired setups.** In heterotic/type II supergravities, the NS–NS three-form $H = dB$ enters via the *torsionful* spin connection $\omega_\pm = \mathring{\omega} \pm \frac{1}{2}\,H$. In slowly varying, parity-even sectors and upon dimensional reduction, gradients of the axion-like fields dual to B act as effective *trace* torsion, yielding ∂_μ of a scalar in the infrared and a projectively invariant combination akin to $\mathcal{T}_\mu$.
2. **'t Hooft naturalness from symmetry.** The projective symmetry together with the shift symmetry $\epsilon \to \epsilon + c$ protects the *gradient-only* appearance of ϵ and suppresses f_ϵ^2 technically: taking $f_\epsilon^2 \to 0$ *enhances* symmetry (exact spurion posture).

These UV *hints* are illustrative only; no UV assumption is needed for the main results.

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